The numerical toolbox: derivatives, integrals, roots

Prognostic Modelling (pea700) · Lecture 3

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Recap of lecture 2

- Gradient, divergence, curl, Laplacian; divergence = net outflow per volume
- Conservation law $\partial u/\partial t + \nabla \cdot \vec{F} = s$, closed by advective and diffusive fluxes
- Classify first: ODE/PDE, order, linearity, elliptic/parabolic/hyperbolic
- Well-posed: right number of conditions, continuous dependence on the data

Today

- Finite differences from the Taylor series, and their truncation error
- Measuring the order of convergence: the basic test of every code
- Why the step cannot be made arbitrarily small: round-off
- Stencils, and what to do at the boundary
- Interpolation, quadrature, root finding

> All tools rest on one idea: the Taylor series. Combine series so that unwanted terms cancel.

Finite differences

The Taylor series

$$u(x+h) = u(x) + h\,u'(x) + \frac{h^2}{2}u''(x) + \frac{h^3}{6}u'''(x) + \frac{h^4}{24}u''''(x) + \ldots$$

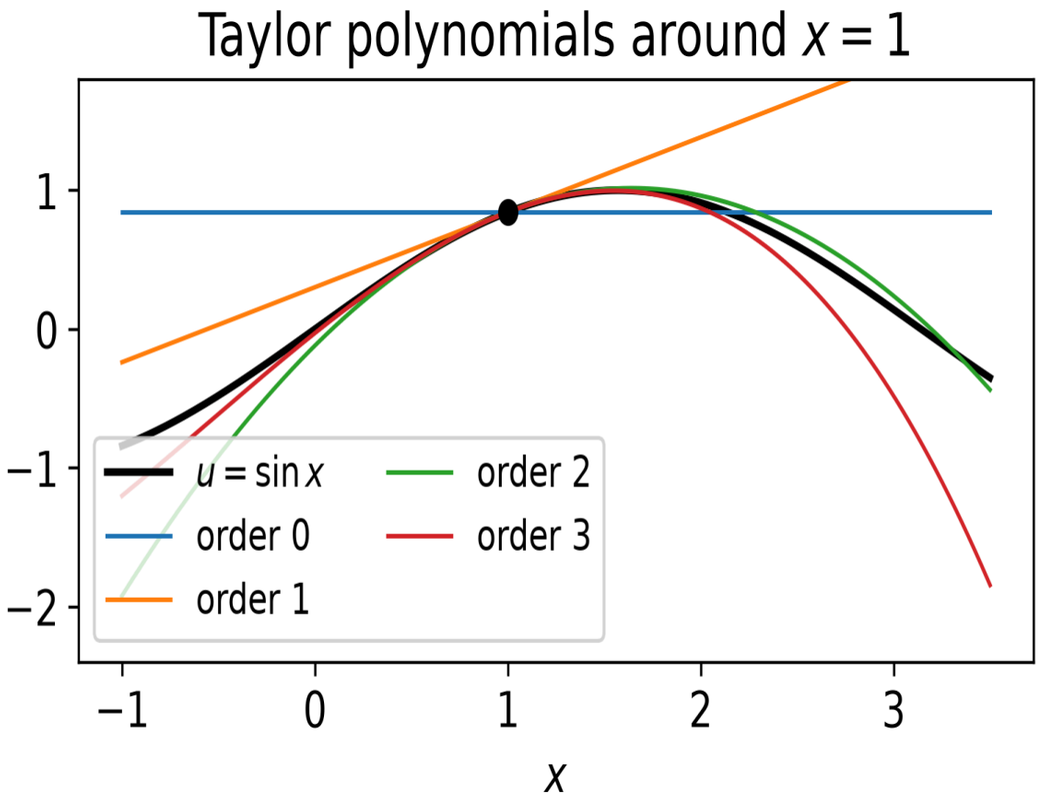
Replace h by $-h$: the **odd** terms change sign, $(-h)^k = -h^k$ for odd k

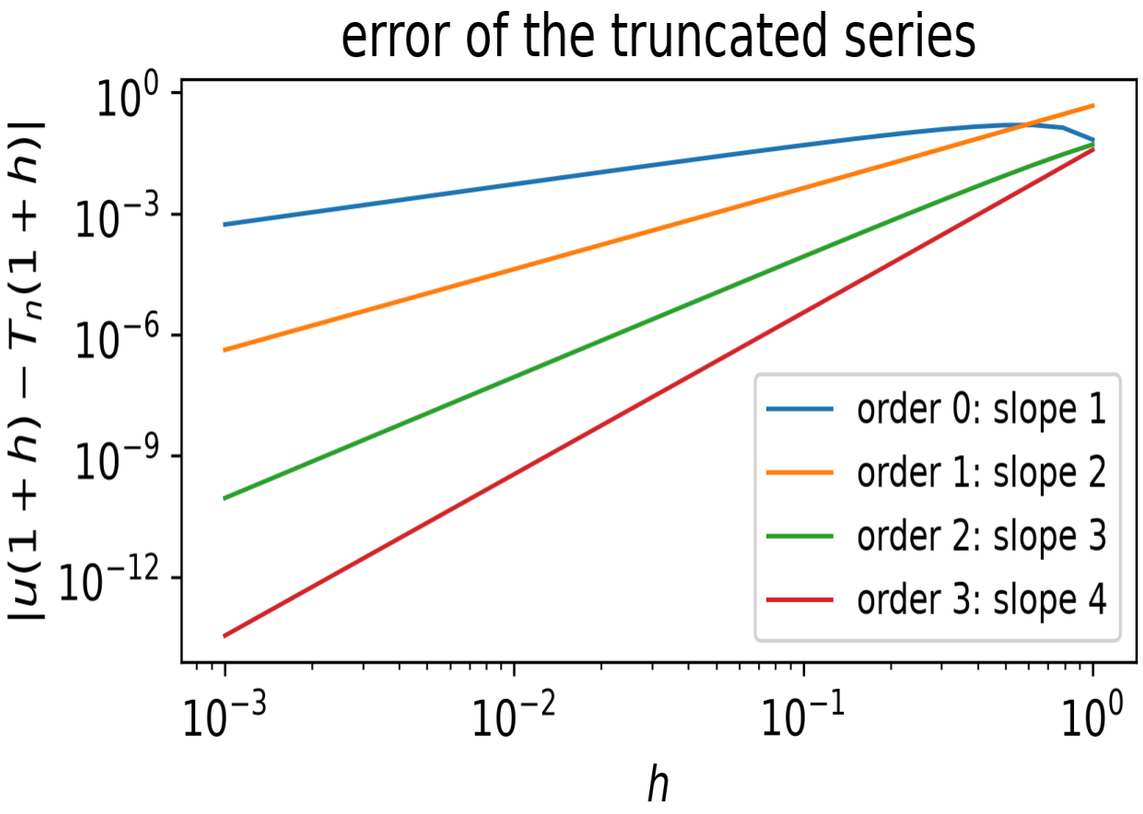
$$u(x-h) = u(x) - h\,u'(x) + \frac{h^2}{2}u''(x) - \frac{h^3}{6}u'''(x) + \frac{h^4}{24}u''''(x) - \ldots$$

Notation: $e = O(h^p)$ if $|e| \leq Ch^p$ for small h, C independent of h.

A method with error $O(h^p)$ is **of order** p: halving h divides the error by about 2^p.

Taylor polynomials





Stopped after h^n: the error behaves like the first neglected term, $\frac{h^{n+1}}{(n+1)!}u^{(n+1)}$, slope $n+1$.

> Every truncation error in this lecture is such a neglected remainder.

Forward difference

Subtract $u(x)$, divide by h:

$$\frac{u(x+h)-u(x)}{h} = u'(x) + \frac{h}{2}u''(x) + \frac{h^2}{6}u'''(x) + \dots$$

Solve for $u'(x)$: every term moves to the other side **with a minus sign**

$$u'(x) = \underbrace{\frac{u(x+h)-u(x)}{h}}_{\text{forward difference}} \underbrace{- \frac{h}{2}u''(x) - \frac{h^2}{6}u'''(x) - \dots}_{\text{truncation error}}$$

First order: the error is proportional to h.

Backward difference

Subtract $u(x-h)$ from $u(x)$, divide by h:

$$\frac{u(x)-u(x-h)}{h} = u'(x) - \frac{h}{2}u''(x) + \frac{h^2}{6}u'''(x) - \dots$$

Solve for $u'(x)$:

$$u'(x) = \underbrace{\frac{u(x)-u(x-h)}{h}}_{\text{backward difference}} \underbrace{+ \frac{h}{2}u''(x) - \frac{h^2}{6}u'''(x) + \dots}_{\text{truncation error}}$$

Also first order, but the leading error $+\frac{h}{2}u''$ has the **opposite sign** to the forward difference.

Central difference

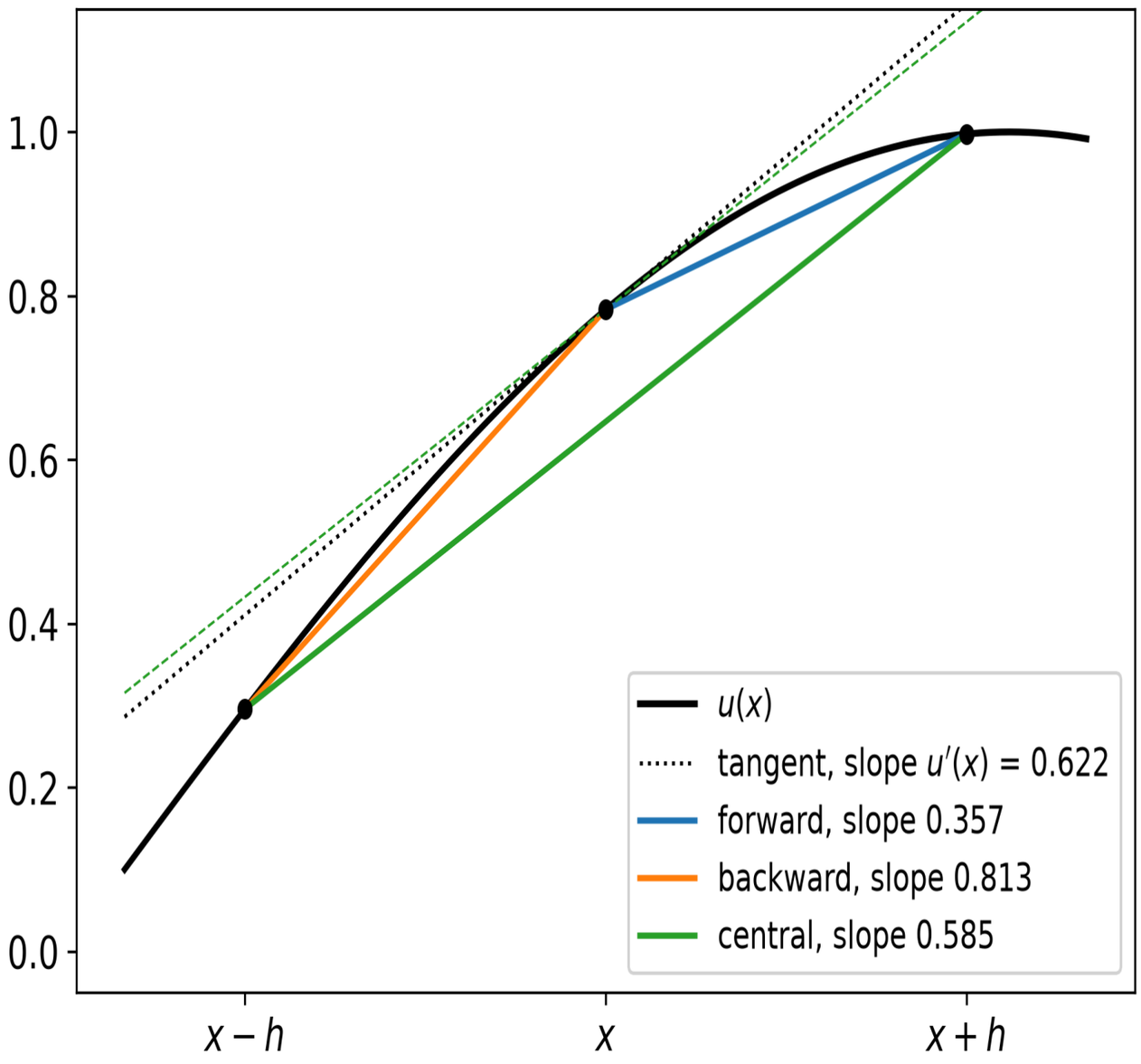
Think first: opposite signs. What happens if we combine them?

Term	in $u(x+h)$	in $u(x-h)$	difference
u	u	u	0
u'	$h\,u'$	$-h\,u'$	$2h\,u'$
u''	$\frac{h^2}{2}u''$	$\frac{h^2}{2}u''$	0
u'''	$\frac{h^3}{6}u'''$	$-\frac{h^3}{6}u'''$	$\frac{h^3}{3}u'''$

Divide by $2h$, solve for $u'(x)$:

$$u'(x) = \frac{u(x+h) - u(x-h)}{2h} - \frac{h^2}{6}u'''(x) - \ldots \qquad \text{second order, same cost}$$

Difference quotients are slopes of chords



- forward and backward chord: one too flat, one too steep, by about $\frac{h}{2}|u''|$
- central chord spans $[x-h, x+h]$ symmetrically: nearly parallel to the tangent (thin dashed)
- parabola, $u''' = 0$: exactly parallel

Think first: for which u is the forward difference exact?

For straight lines, $u = a + bx$: then $u'' = 0$ and every term of the truncation error vanishes. The chord is the tangent, for every h.

Second derivative

Add the two series: the odd terms cancel

$$u(x+h) + u(x-h) = 2u(x) + h^2 u''(x) + \frac{h^4}{12} u''''(x) + \dots$$

Subtract $2u(x)$, divide by h^2, solve for $u''(x)$:

$$u''(x) = \frac{u(x+h) - 2u(x) + u(x-h)}{h^2} - \frac{h^2}{12} u''''(x) - \dots$$

Flux form: difference of two first derivatives at the half points

$$u''(x) \approx \frac{1}{h} \left[\underbrace{\frac{u(x+h) - u(x)}{h}}_{u'(x+h/2)} - \underbrace{\frac{u(x) - u(x-h)}{h}}_{u'(x-h/2)} \right]$$

The core of every diffusion model; with variable D it becomes $D_{i+1/2}$ and $D_{i-1/2}$.

Measuring the order of convergence

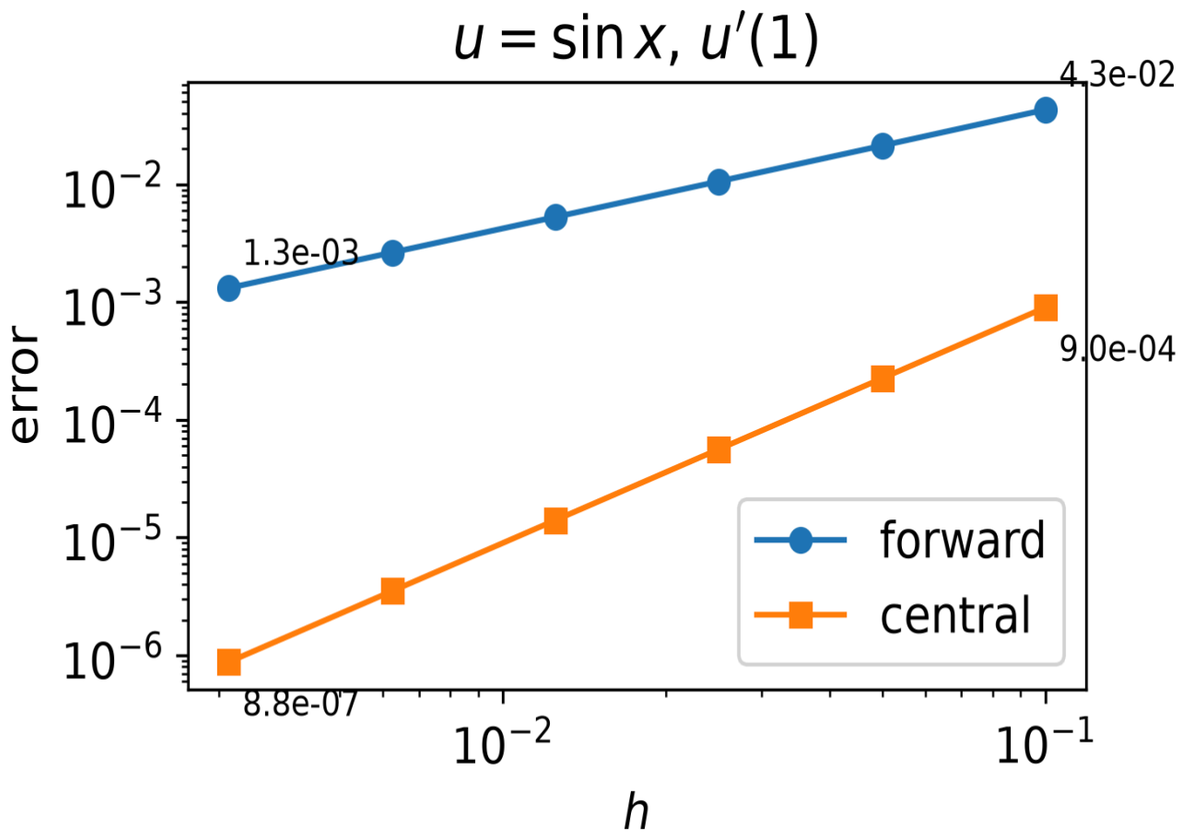
$$e(h) \approx Ch^p \;\Rightarrow\; \log e = \log C + p \log h$$

Straight line with slope p on log–log axes.

From two steps, C drops out:

$$p = \frac{\log\left(e(h_1)/e(h_2)\right)}{\log\left(h_1/h_2\right)}$$

Halving h: $p = \log_2$ of the error ratio.
Factor 4 = second order.



> Observed order = theoretical order is **the** test of a code. A bug almost always lowers the order.

Truncation versus round-off

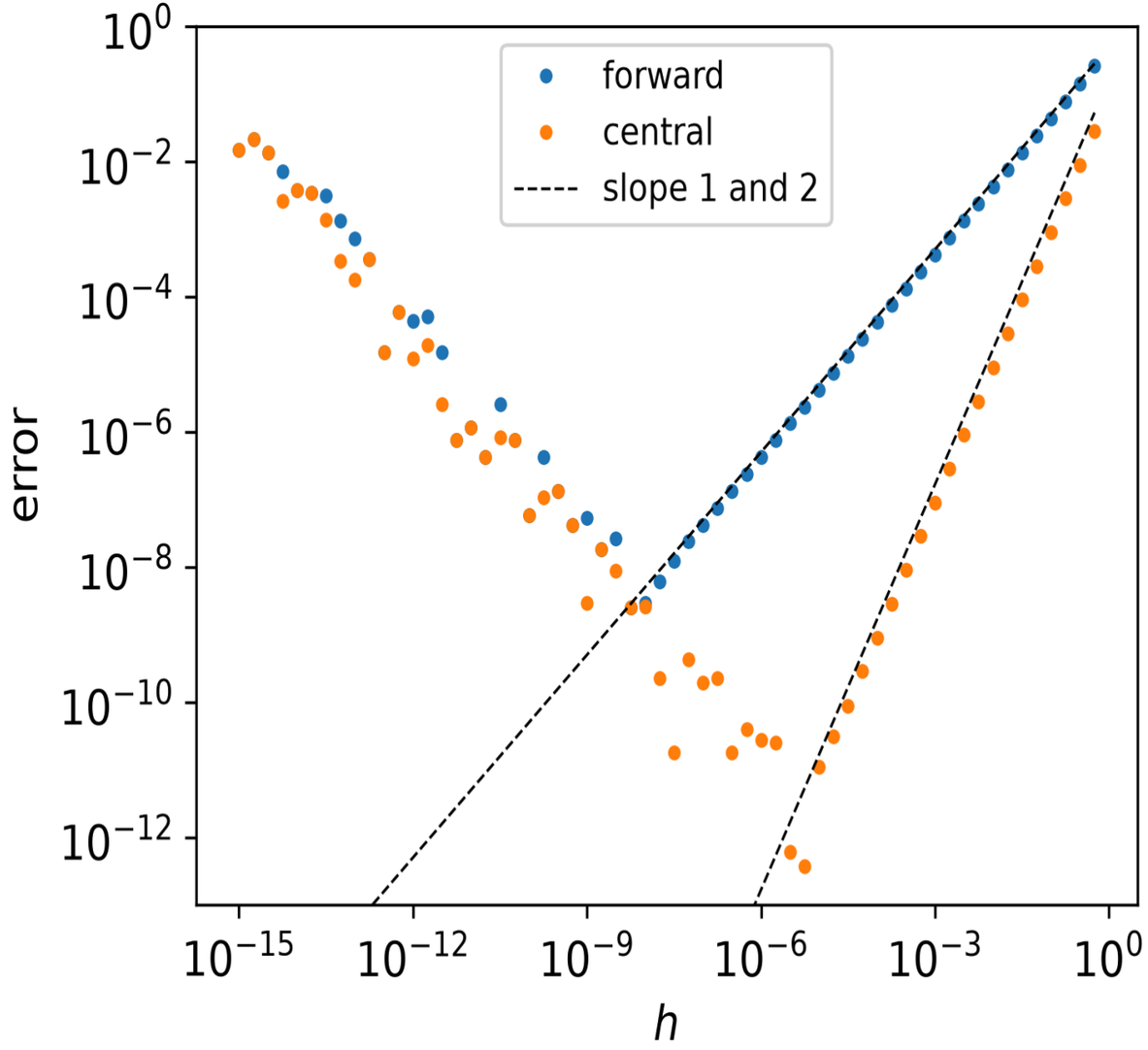
Stored values carry relative errors $|\delta| \le \varepsilon$:

$$u(x+h)(1+\delta_2) - u(x)(1+\delta_1)$$

Numerator wrong by up to $2\varepsilon|u|$; divided by h:

$$e(h) = \underbrace{\frac{h}{2}|u''|}_{\text{truncation}} + \underbrace{\frac{2\varepsilon|u|}{h}}_{\text{round-off}}$$

Think first: where is the minimum?



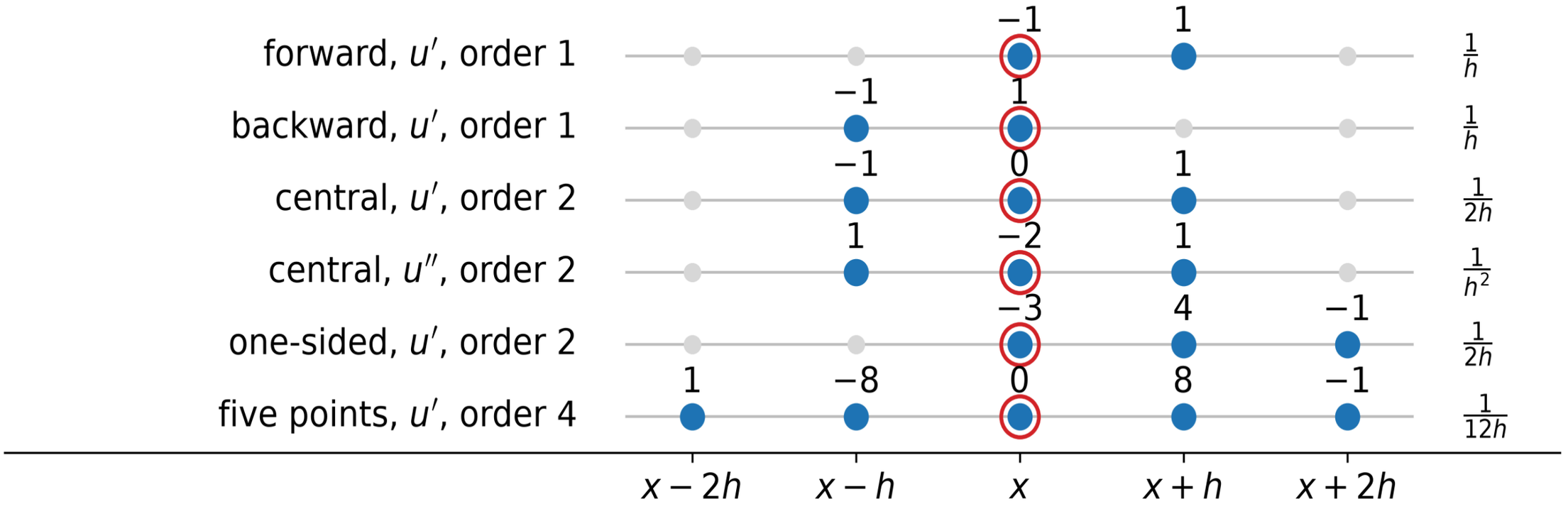
The optimal step

$$\frac{\mathrm{d}e}{\mathrm{d}h} = \frac{|u''|}{2} - \frac{2\varepsilon|u|}{h^2} = 0 \quad \Longrightarrow \quad h_{\mathrm{opt}} = 2\sqrt{\frac{\varepsilon|u|}{|u''|}}, \qquad e(h_{\mathrm{opt}}) = 2\sqrt{\varepsilon\,|u|\,|u''|}$$

	h_{opt}	**smallest error**	**digits lost**
forward, $O(h)$	$\sim \sqrt{\varepsilon} \approx 10^{-8}$	$\sim 10^{-8}$	half of 16
central, $O(h^2)$	$\sim \varepsilon^{1/3} \approx 10^{-5}$	$\sim \varepsilon^{2/3} \approx 10^{-11}$	about 5

> Higher order: smaller error at the same step and less sensitive to round-off. In a convergence test, choose h neither so large that higher terms disturb the slope nor so small that round-off dominates.

Stencils



- **Boundary**: the central stencil needs a point outside the domain. Instead: one-sided $(-3, 4, -1)/2h$, second order (exercise sheet), or a ghost point
- **Next lecture**: one stencil per grid point is one row of a matrix; $1, -2, 1$ gives a tridiagonal matrix

Linear interpolation and its error

$$p(x) = u_j + \frac{u_{j+1} - u_j}{h}(x - x_j), \qquad \text{midpoint: } \tfrac{1}{2}(u_j + u_{j+1})$$

Needed constantly: values between grid points, regridding, departure points of air parcels.

Error of a polynomial of degree n through $n+1$ points:

$$u(x) - p(x) = \frac{u^{(n+1)}(\xi)}{(n+1)!} \prod_{j=0}^{n} (x - x_j)$$

$n = 1, t = x - x_j$: the product $t(t-h)$ is extreme at $t = h/2$, value $-h^2/4$

$$|u(x) - p(x)| \le \frac{h^2}{8} \max |u''| \qquad \text{second order, exact without curvature}$$

The Runge phenomenon

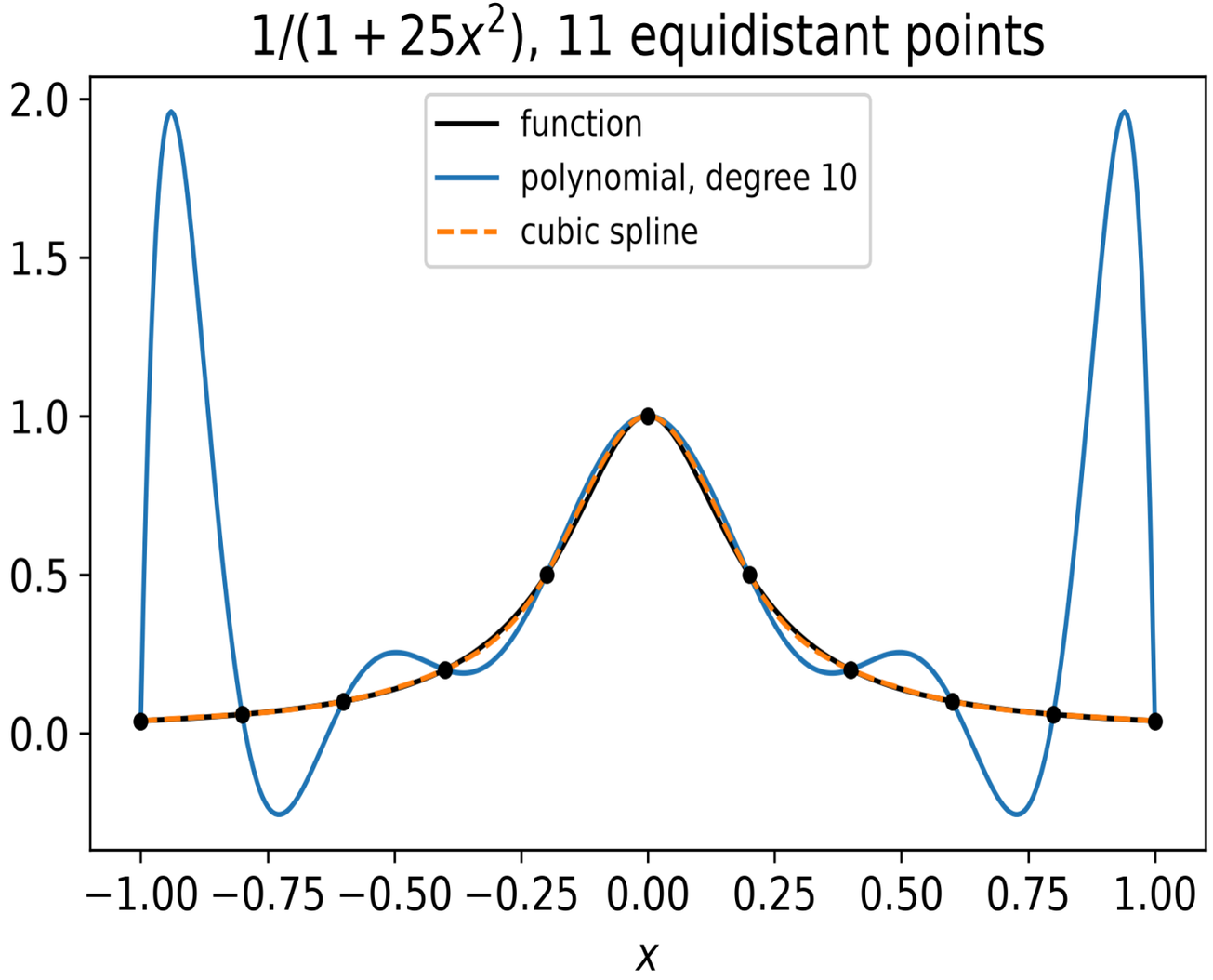
Think first: more points, higher degree, better fit?

Not on equidistant points: $\prod(x - x_j)$ is large near the ends, and $u^{(n+1)}$ can grow faster than $(n+1)!$.

Remedies:

- piecewise, low degree (**splines**)
- cluster points at the ends (Chebyshev); a model grid rarely allows this

> Refine the grid with a method of moderate order.



Quadrature

Midpoint and trapezoidal rule

One subinterval of width h, midpoint m. Expand $f(m+s)$, integrate from $-h/2$ to $h/2$; odd powers of s integrate to zero:

$$\int_{-h/2}^{h/2} s^2\,\mathrm{d}s = \frac{2}{3}\left(\frac{h}{2}\right)^3 = \frac{h^3}{12} \quad\Longrightarrow\quad \int f\,\mathrm{d}x = \underbrace{h\,f(m)}_{\text{midpoint}} + \frac{h^3}{24}f''(m) + O(h^5)$$

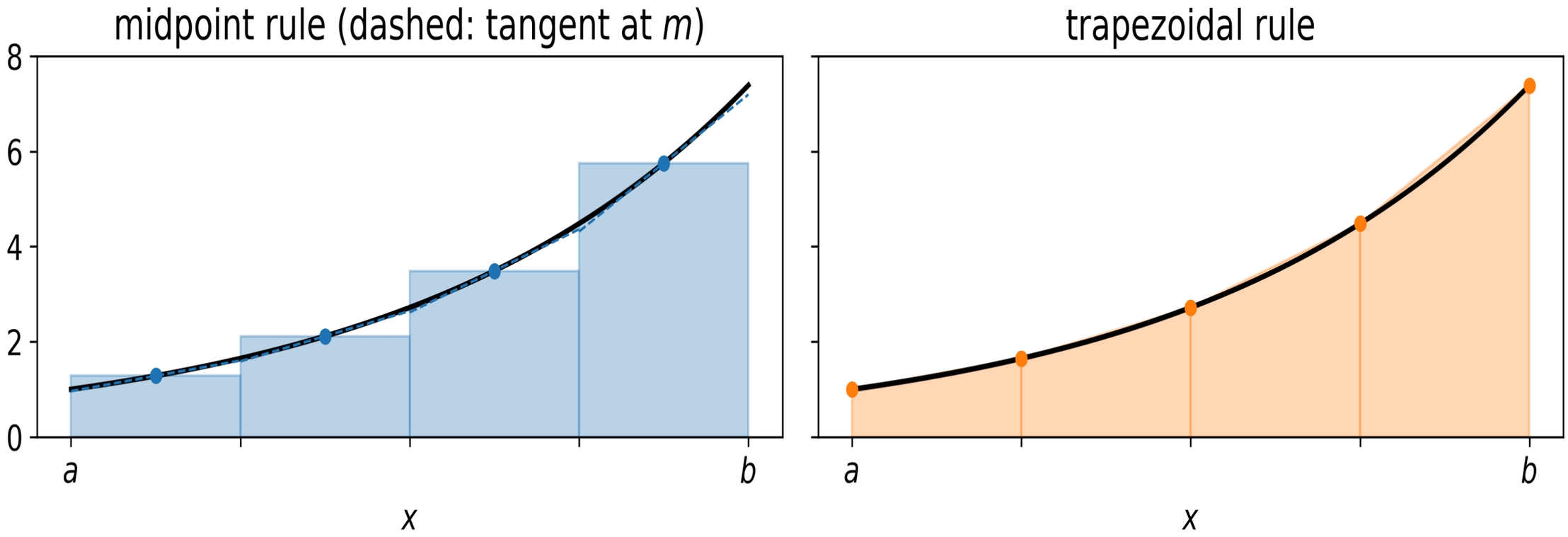
End points: $f(m-\frac{h}{2}) + f(m+\frac{h}{2}) = 2f(m) + \frac{h^2}{4}f''(m) + O(h^4)$, so

$$\underbrace{\frac{h}{2}\left[f(m-\tfrac{h}{2}) + f(m+\tfrac{h}{2})\right]}_{\text{trapezoidal}} = hf(m) + \frac{h^3}{8}f'' = \int f\,\mathrm{d}x \underbrace{-\frac{h^3}{24}f'' + \frac{h^3}{8}f''}_{+\frac{h^3}{12}f''}$$

Solved for the integral: $\int f\,\mathrm{d}x = \text{trapezoidal} - \frac{h^3}{12}f''(m) + O(h^5)$.

Twice the midpoint error, **opposite sign**.

Which sign? The picture



Convex f ($f'' > 0$): chords lie above the curve, so **trapezoidal too large.** The rectangle has the area of the trapezoid under the **tangent** at m, $\int_{-h/2}^{h/2} \Big(f(m) + s f'(m)\Big)\mathrm{d}s = h f(m)$, and tangents lie below: **midpoint too small**.

Composite rules and Simpson

$N = (b-a)/h$ subintervals: local $O(h^3)$ adds up to $O(h^2)$

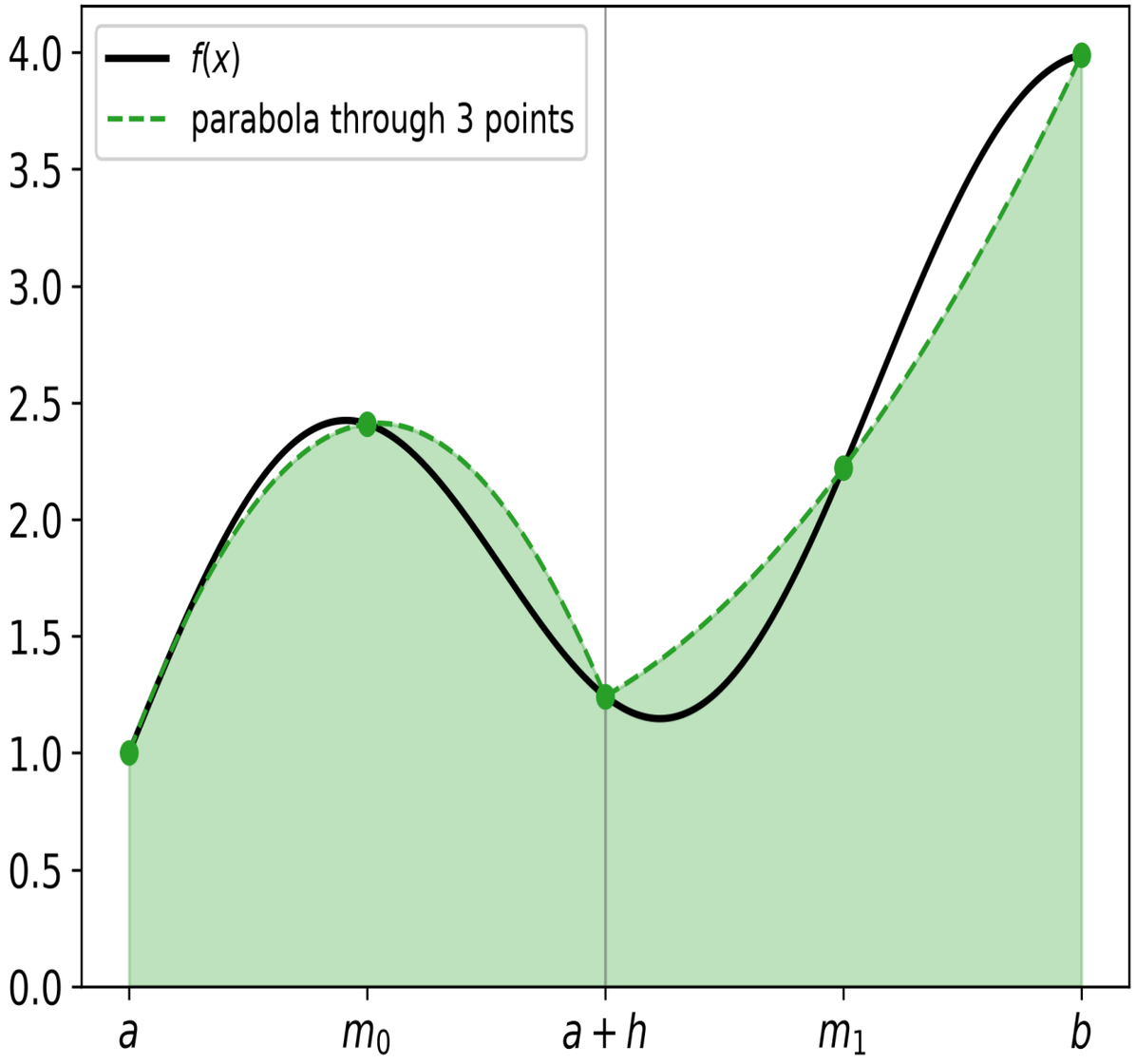
$$\text{midpoint: } \frac{h^2}{24}\big[f'(b) - f'(a)\big], \qquad \text{trapezoidal: } -\frac{h^2}{12}\big[f'(b) - f'(a)\big]$$

Periodic f over one period: $f'(b) = f'(a)$, the trapezoidal rule is extremely accurate.

Simpson: $\frac{2}{3}$ midpoint $+\frac{1}{3}$ trapezoidal, the h^3 terms cancel: $-\frac{2}{3}\cdot\frac{1}{24} + \frac{1}{3}\cdot\frac{1}{12} = 0$

$$\int_{m-h/2}^{m+h/2} f\,\mathrm{d}x \approx \frac{h}{6}\left[f(m-\tfrac{h}{2}) + 4f(m) + f(m+\tfrac{h}{2})\right] \qquad \text{fourth order}$$

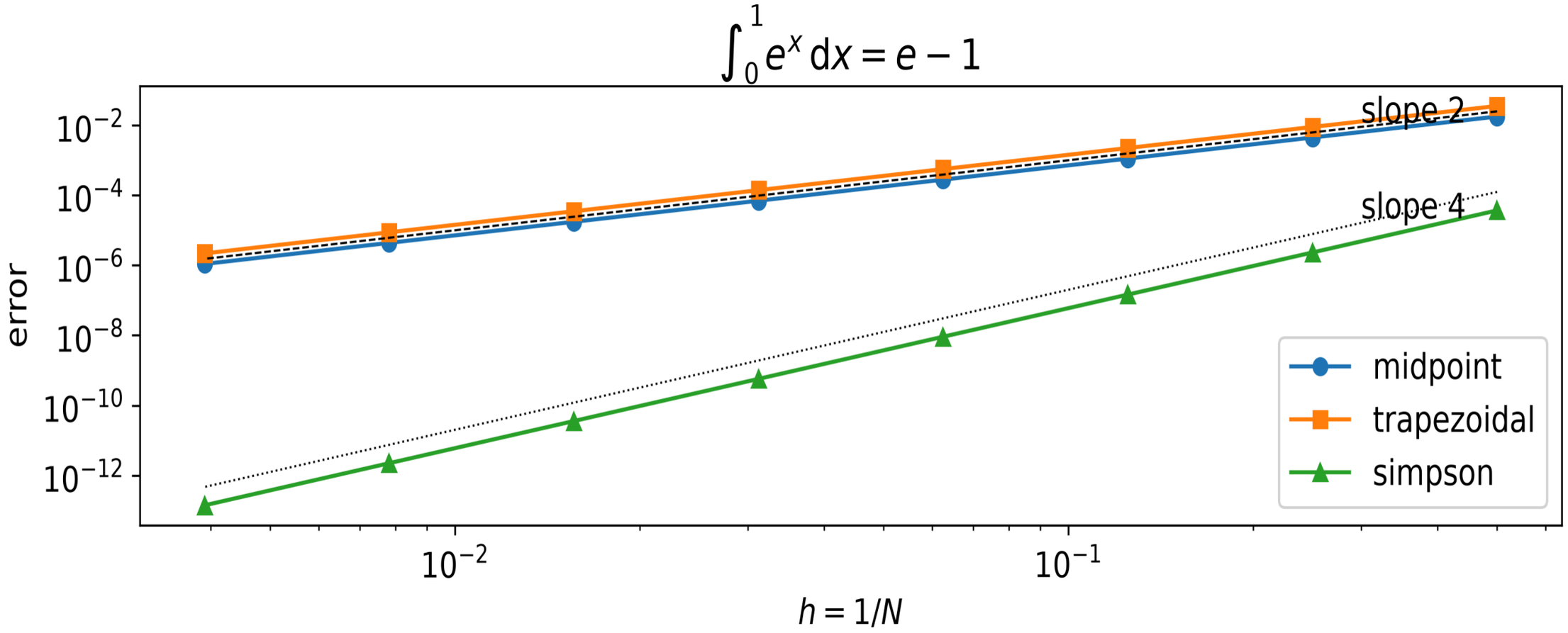
Simpson integrates a parabola



- error terms contain only f'''' and higher: exact for **cubics**
- parabola p through the 3 points: Simpson $(p) = \int p$
- Simpson uses only the 3 values, where $p = f$:

$$\text{Simpson}(f) = \text{Simpson}(p) = \int p \, \mathrm{d}x$$

Three rules, one test



In models: every time stepping scheme is a quadrature rule for $\int_t^{t+\Delta t} f \, \mathrm{d}t$ (left rectangle = forward Euler); finite volumes integrate fluxes over faces; finite elements use Gauss quadrature.

Root finding

How warm does the ground get?

Nonlinear equations everywhere: implicit steps, saturation temperature, water table. Example: dry ground at noon.

$$\underbrace{\sigma T^4}_{\text{emission}} + \underbrace{h\,(T - T_a)}_{\text{sensible heat}} = \underbrace{Q}_{\text{absorbed}} \qquad \Longrightarrow \qquad f(T) = \sigma T^4 + h\,(T - T_a) - Q = 0$$

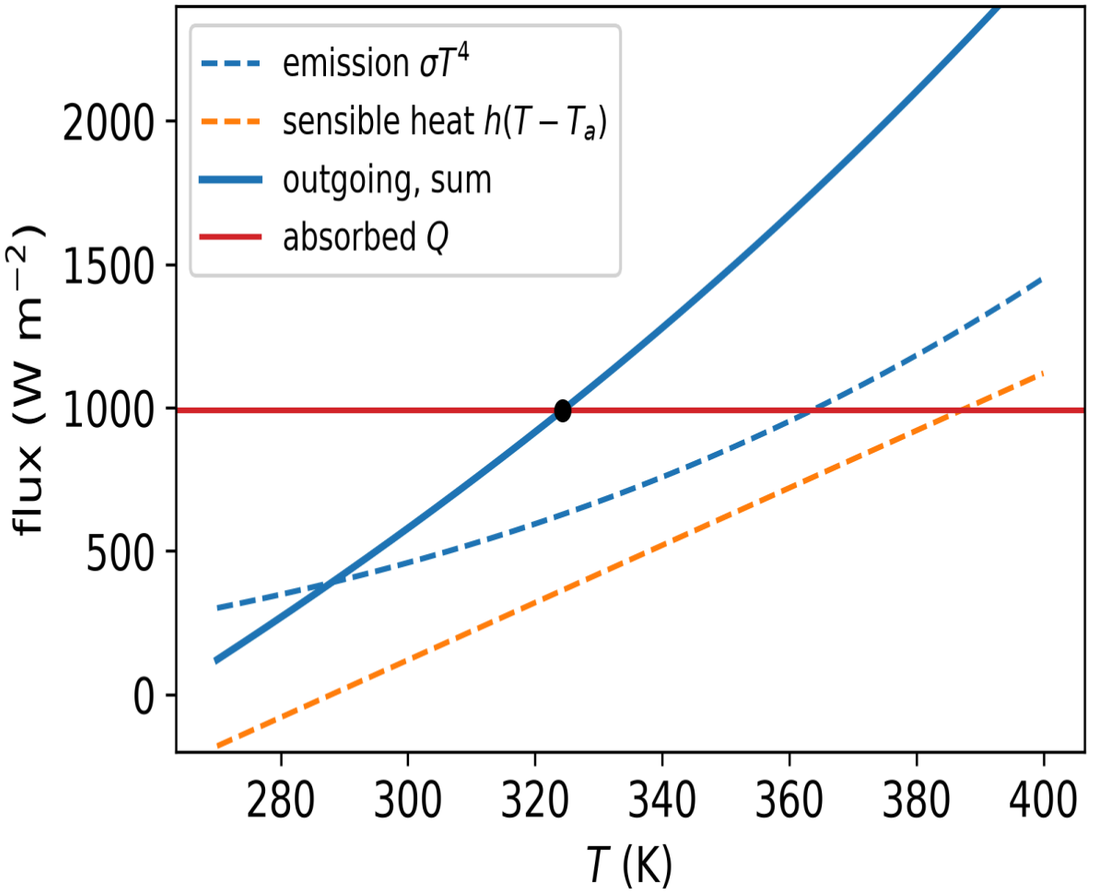
$\sigma = 5.67 \cdot 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$, $h = 10\,\mathrm{W\,m^{-2}\,K^{-1}}$, $T_a = 288\,\mathrm{K}$, $Q = 990\,\mathrm{W\,m^{-2}}$.

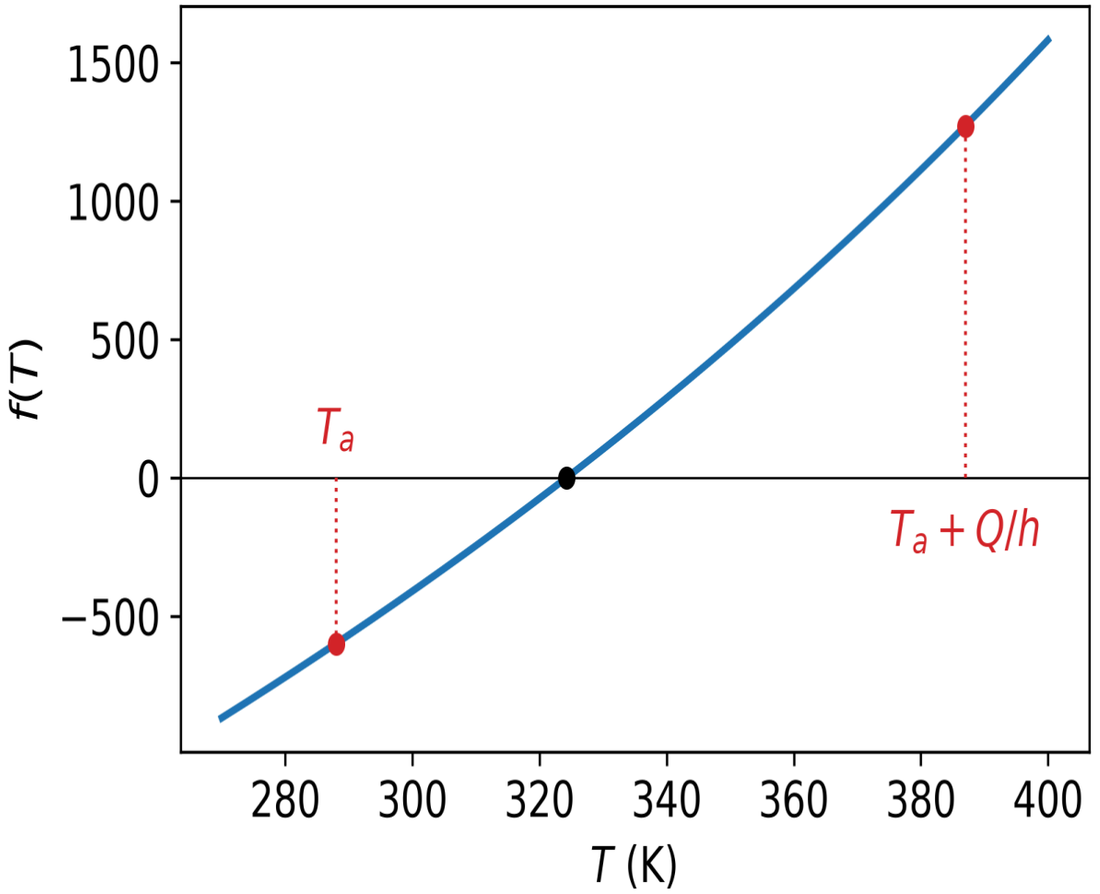
Think first: between which two temperatures must the root lie?

- $T = T_a$: no sensible heat, emission $390 < Q$, so $f = -600 < 0$
- $T = T_a + Q/h = 387\,\mathrm{K}$: sensible heat alone is Q, so $f = \sigma T^4 > 0$

Fourth degree, no usable formula: solve iteratively, in every grid cell and time step. $f'(T) = 4\sigma T^3 + h > 0$.

The balance and the bracket





Root $T = 324.3\,\mathrm{K} \approx 51\,°\mathrm{C}$, unique because the outgoing energy grows with T. Caution: a sign change needs a **continuous** f; $1/x$ changes sign at its pole.

Bisection

Sign change on $[a, b] \Rightarrow$ root inside. Halve, keep the half with the sign change.

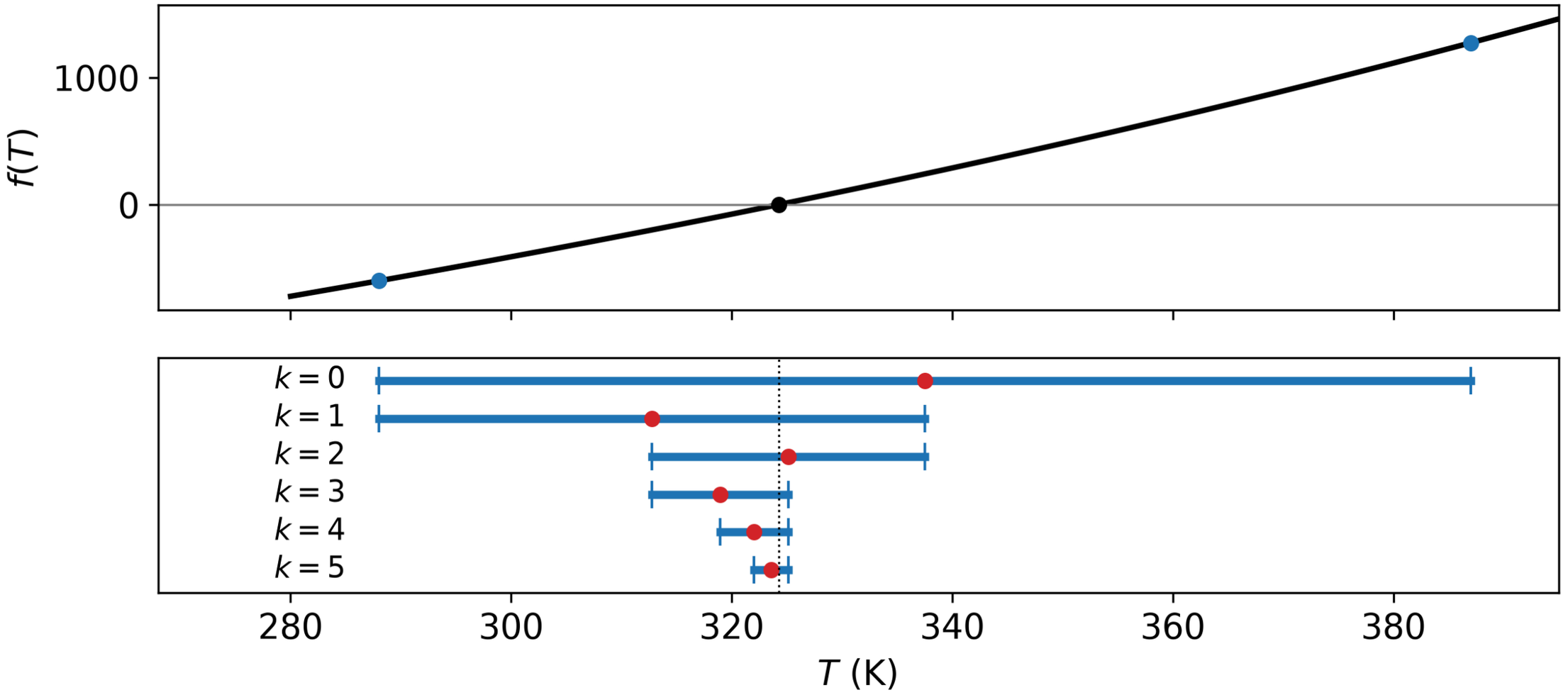
$$e_k \leq \frac{b-a}{2^{k+1}} \qquad \Longrightarrow \qquad k \geq \log_2 \frac{b-a}{\tau} - 1$$

$b - a = 99$ K, $\tau = 10^{-12}$ K: 46 iterations. One binary digit per iteration, 3.3 per decimal digit.

Linear convergence: the error bound shrinks by a fixed factor, $\frac{1}{2}$.

- cannot fail (if f is continuous), but slow
- does not extend to systems

Bisection: the intervals



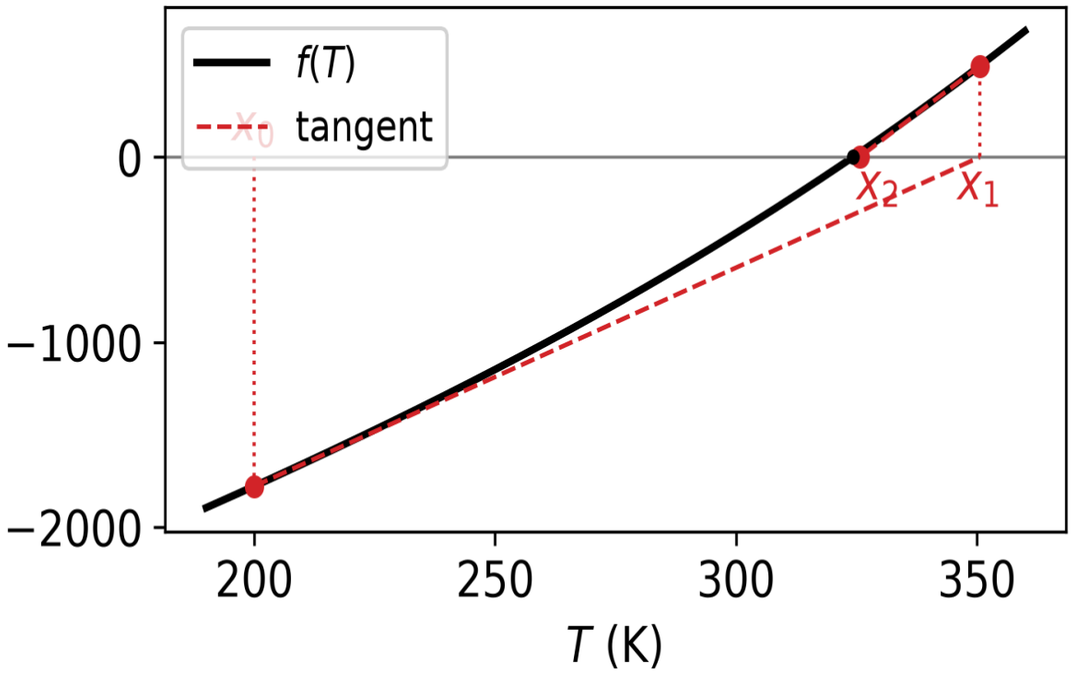
Each interval is half of the previous one and contains the root (dotted).

Newton–Raphson

Keep two Taylor terms (the tangent) and set to zero:

$$f(x_k + \delta) \approx f(x_k) + \delta\, f'(x_k) = 0$$

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$



From $x_0 = 200$ K: 350.6, 325.7 K. f convex: tangent below the curve, so x_1 lands **right** of the root.

Error $e_k = x_k - r$, so the step from x_k to the root is $r - x_k = -e_k$. Taylor at x_k, evaluated at r:

$$0 = f(r) = f(x_k) - e_k\, f'(x_k) + \frac{e_k^2}{2} f''(\xi)$$

Quadratic convergence

Divide by $f'(x_k)$ and move $-e_k$ and f/f' to the left (both change sign):

$$e_k - \frac{f(x_k)}{f'(x_k)} = \frac{f''(\xi)}{2f'(x_k)}\, e_k^2$$

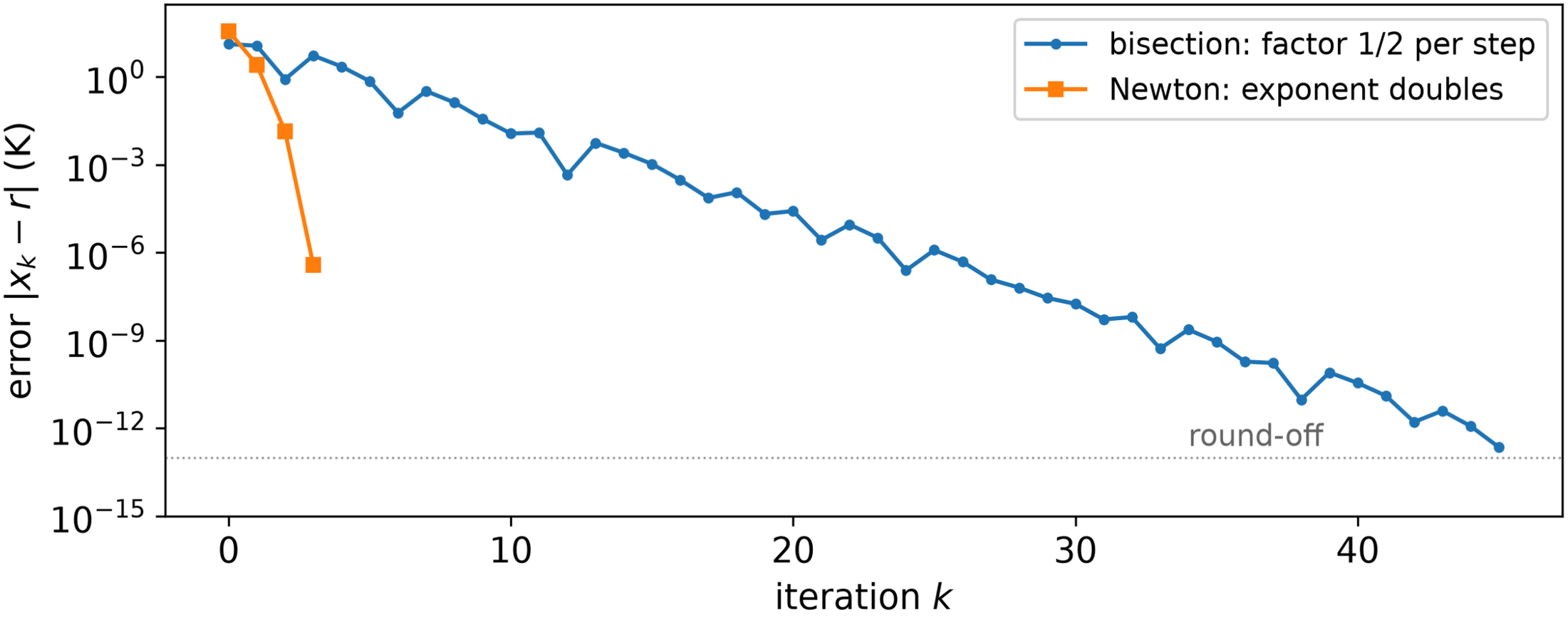
Left side: $x_k - r - \frac{f(x_k)}{f'(x_k)} = x_{k+1} - r = e_{k+1}$. Hence

$$\boxed{e_{k+1} \approx \frac{f''(r)}{2f'(r)}\, e_k^2}$$

From $T_a = 288$ K the errors are 2.6 K $\to$ 0.014 K $\to$ $4 \cdot 10^{-7}$ K: correct digits **double** per iteration.

- needs f', and a good start (where $f' \approx 0$ the tangent throws x far away)
- double root, $f'(r) = 0$: only linear
- practice: bisection to get close, then Newton (`scipy.optimize.root_scalar`, `brentq`)

Linear versus quadratic convergence



Bisection: a straight line on a log axis (factor $\frac{1}{2}$). Newton: the curve bends down ever more steeply, round-off after 4 iterations.

Newton for systems

$\vec{F}(\vec{x}) = 0$, N equations. First-order Taylor with the **Jacobian** $J_{ij} = \partial F_i / \partial x_j$:

$$\vec{F}(\vec{x}_k + \delta\vec{x}) \approx \vec{F}(\vec{x}_k) + J(\vec{x}_k)\,\delta\vec{x} = 0 \qquad \Longrightarrow \qquad J(\vec{x}_k)\,\delta\vec{x} = -\vec{F}(\vec{x}_k)$$

Circle and hyperbola, start $(2, 0.5)$:

$$\vec{F} = \begin{pmatrix} x^2 + y^2 - 4 \\ xy - 1 \end{pmatrix}, \qquad J = \begin{pmatrix} 2x & 2y \\ y & x \end{pmatrix}$$

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error in x:  1.5e-03   1.1e-06   8.0e-13   2.2e-16
```

> A nonlinear problem becomes a sequence of **linear systems**: how the Richards equation is solved. Next lecture: linear algebra.

Summary

- Finite differences and their truncation errors follow from the Taylor series; forward/backward are first order, central first and second derivatives second order.
- Measure the order on a log–log plot or from error ratios: the basic verification test.
- Round-off limits finite differences: the best step is about $\sqrt{\varepsilon}$ (forward), $\varepsilon^{1/3}$ (central).
- A stencil is the set of points of a formula; at boundaries one-sided formulas or ghost points replace the central stencil.
- Linear interpolation, midpoint and trapezoidal rules are second order; Simpson fourth; high-degree polynomials on equidistant points oscillate.
- Bisection: linear and safe. Newton: quadratic near a simple root, and works for systems via the Jacobian.

Lab and next lecture

- **Lab**: start Exercise 3 (truncation and round-off, one-sided difference, quadrature, root finding), due date
- Start your `toolbox.py`: difference formulas, quadrature rules, bisection, Newton
- **Reading**: script, unit 3
- **Next lecture**: the numerical toolbox II: matrices and waves